

DBMS Class Exercise 1 - Submitted by Abhishek Vincent

1. Table Design – Foreign Keys and Normalization

Table: Department

Description: Used to store Department Information

Column Name	Description
DeptNo	Department ID
DeptName	Department Name
hod_name	Head of Department
location	Office Location

Table: Employee

Description: Used to store Employee Information

Column Name	Description
EmpNo	Employee ID (must start with 'E')
EmpName	Employee Name
DeptNo	Department ID (references Department)
Mobile	Employee Contact Number
Hire_date	Date of Joining
Job	Job Title / Designation
Edlevel	Education Level
gender	Gender ('M' or 'F')
dob	Date of Birth
salary	Basic Salary (must be > 0)
bonus	Bonus Amount
comm	Commission Amount

Table: Project

Description: Used to store details of projects in the organisation

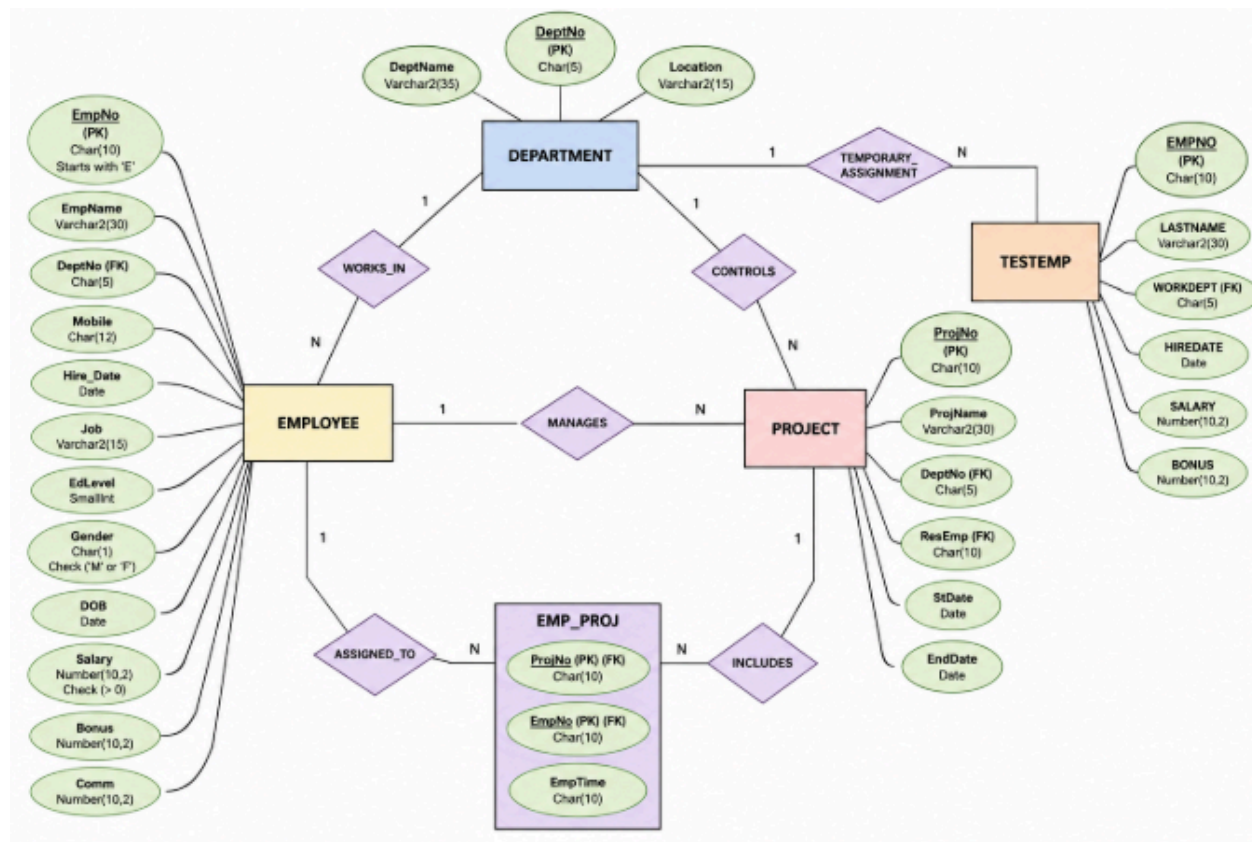
Column Name	Description
ProjNo	Project ID
ProjName	Project Name
DeptNo	Department ID (references Department)
ResEmp	Responsible Employee (references Employee)
StDate	Project Start Date
EndDate	Project End Date

Table: Emp_Proj

Description: Used to store details of employee activity for a project

Column Name	Description
ProjNo	Project ID (references Project)
EmpNo	Employee ID (references Employee)
EmpTime	Time Spent by Employee on Project

E-R Diagram



2. Practice SQL

2.1 Table Creation and Alteration

Create the tables based on the above description:

```
drop table Emp_Proj;
drop table project;
drop table employee;
drop table department;

create table department(
DeptNo char(5) constraint dept_pkey primary key not null,
DeptName varchar2(35),
hod_name varchar2(25),
location varchar2(50)
);

create table employee(
EmpNo char(10) constraint employee_pkey primary key check (EmpNo like 'E%'),
EmpName varchar2(30),
DeptNo char(5) references department(DeptNo),
Mobile char(12),
Hire_date date,
Job varchar2(15),
Edlevel smallint,
```

```

gender char(1) constraint employee_gender_check check(gender in ('M','F')),
dob date,
salary number(10,2) constraint employee_salary_check check ( salary > 0 ),
bonus number(10,2),
comm number(10,2)
);

create table project(
ProjNo char(10) constraint project_pkey primary key not null,
ProjName varchar2(30) ,
DeptNo char(5) references department(DeptNo),
ResEmp char(10) references employee(EmpNo),
StDate date,
EndDate date
);

create table Emp_Proj(
ProjNo char(10) references project(ProjNo),
EmpNo char(10) references employee(EmpNo),
EmpTime char(10)
);

```

Add a column called 'Address' of data type Varchar and size 50 to the Employee table:

```

alter table employee
add Address varchar2(50);

```

```

SQL> alter table employee
      2  add Address varchar2(50);

Table altered.

SQL>

```

Drop the above created column:

```

alter table employee
drop column Address;

```

```

SQL> alter table employee
      2  drop column Address;

Table altered.

SQL>

```

Change the size of the PhoneNo (Mobile) column in Employee to 10:

```

alter table employee
modify Mobile char(10);

```

```
SQL> alter table employee
  2  modify Mobile char(10);

Table altered.

SQL> |
```

Change the name of the column ResEmp in Project table to RespEmp:

```
alter table project
rename column ResEmp to RespEmp;
```

```
SQL> alter table project
  2  rename column ResEmp to RespEmp;

Table altered.

SQL> |
```

Destroy the table Emp_Proj and then recreate it:

```
drop table Emp_Proj;

create table Emp_Proj(
ProjNo char(10) references project(ProjNo),
EmpNo char(10) references employee(EmpNo),
EmpTime char(10)
);
```

```
SQL> drop table Emp_Proj;

Table dropped.

SQL>
SQL> create table Emp_Proj(
  2  ProjNo char(10) references project(ProjNo),
  3  EmpNo char(10) references employee(EmpNo),
  4  EmpTime char(10)
  5  );

Table created.

SQL> |
```

Add check constraints — Hire date > DOB; Project start date < end date:

```
alter table employee
add constraint chk_hiredate_dob check (Hire_date > dob);

alter table project
add constraint chk_stdate_enddate check (StDate < EndDate);
```

```
SQL> alter table employee
      2  add constraint chk_hiredate_dob check (Hire_date > dob);

Table altered.

SQL>
SQL> alter table project
      2  add constraint chk_stdate_enddate check (StDate < EndDate);

Table altered.

SQL> |
```

3. Practice SQL – Data Manipulation Language (DML)

3.1 Row Insertion, Deletion and Updating

Insert the data for the tables with data given in the excel file 'Dataset.xls':

```
-- Data inserted for Department, Employee, Project and Emp_Proj
-- tables using the records provided in 'Dataset.xls'.
-- Example (repeated for every row in the dataset):

insert into department (DeptNo, DeptName, hod_name, location)
values ('D01', 'Development', 'John Miller', 'Block A');

insert into employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job,
Edlevel, gender, dob, salary, bonus, comm)
values ('E0100', 'HAAS', 'D01', '9876543210', DATE '2005-01-01',
'Manager', 18, 'F', DATE '1975-06-15', 52750, 1000, 4220);

INSERT INTO department (DeptNo, DeptName, location) VALUES ('A00', 'COMPUTER SERVICE
DIVISION', 'KOCHI');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('B01', 'PLANNING',
'BANGALORE');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('C01', 'INFORMATION CENTER',
'KOCHI');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('D01', 'DEVELOPMENT CENTER',
'KOCHI');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('D11', 'MANUFACTURING
SYSTEMS', 'CHENNAI');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('D21', 'ADMINISTRATION
SYSTEMS', 'BANGALORE');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('E01', 'SUPPORT SERVICES',
'CHENNAI');
```

```

INSERT INTO department (DeptNo, DeptName, location) VALUES ('E11', 'OPERATIONS',
'BANGALORE');
INSERT INTO department (DeptNo, DeptName, location) VALUES ('E21', 'SOFTWARE SUPPORT',
'KOCHI');

INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0010', 'ARNOLD LEONARD AMON', 'A00', '9445876534', DATE '2000-12-01',
'PRESIDENT', 18, 'M', DATE '1967-06-19', 152750, 10000, 14220);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0020', 'PHILIP VINCENT', 'B01', '9445976535', DATE '2002-09-10', 'MANAGER', 18,
'M', DATE '1976-12-07', 41250, 4800, 3300);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0030', 'SHILVY K K', 'C01', '9445976538', DATE '2004-06-03', 'MANAGER', 20,
'F', DATE '1978-03-15', 38250, 4800, 3060);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0050', 'ALFERIN LUIZ', 'E01', '9445976541', DATE '2006-03-29', 'MANAGER', 16,
'F', DATE '1982-07-20', 40175, 5800, 3214);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0060', 'OSHINA ANTONY', 'D11', '9445976544', DATE '2008-01-06', 'MANAGER', 16,
'F', DATE '1987-05-12', 32250, 4600, 2580);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0070', 'BINCY PAUL', 'D21', '9445976547', DATE '2009-10-15', 'MANAGER', 16,
'F', DATE '1980-03-30', 36170, 3700, 2893);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0090', 'ANAMIKA PAUL', 'E11', '9445976550', DATE '2011-07-25', 'MANAGER', 16,
'F', DATE '1978-03-19', 29750, 5600, 2380);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0100', 'ANEFESH DENNY', 'E21', '9445976553', DATE '2013-05-03', 'MANAGER', 14,
'M', DATE '1993-10-22', 26150, 5500, 2092);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0110', 'DONA ANICE SIBY', 'A00', '9445976556', DATE '2015-02-10', 'SALESREP',
19, 'F', DATE '1992-09-09', 46500, 1900, 3720);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0120', 'JUNAID K V', 'A00', '9445976559', DATE '2016-11-19', 'CLERK', 14, 'M',
DATE '1997-08-22', 29250, 1600, 2340);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0130', 'CHRISTEENA THOMAS', 'C01', '9445976562', DATE '1999-12-02', 'ANALYST',
16, 'F', DATE '1982-07-20', 23800, 1500, 1904);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0140', 'JEFFIN DOMINIC', 'C01', '9445976565', DATE '2003-09-10', 'ANALYST', 18,
'F', DATE '1982-11-23', 28420, 2600, 2274);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0150', 'JEWEL BIJOY', 'D11', '9445976568', DATE '2007-06-19', 'DESIGNER', 16,
'M', DATE '1984-03-21', 25280, 2500, 2022);

```

```

INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0160', 'MELLOW REEBA THOMAS', 'D11', '9466976556', DATE '2011-03-28',
'DESIGNER', 17, 'F', DATE '1982-02-14', 22250, 2400, 1780);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0170', 'JOHN VARGHESE', 'D11', '9466976557', DATE '2015-01-04', 'DESIGNER', 16,
'M', DATE '1987-11-10', 24680, 2500, 1974);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0180', 'ASHREENA HASSAN', 'D11', '9466976558', DATE '2009-10-15', 'DESIGNER',
17, 'F', DATE '1985-12-26', 21340, 2500, 1707);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0190', 'VISHAK VIJAYAKUMAR', 'D11', '9466976559', DATE '2011-07-25',
'DESIGNER', 16, 'M', DATE '1989-04-29', 20450, 2400, 1636);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0200', 'CORRINNE ELIZABETH RODRIGUES', 'D11', '9466976560', DATE '2013-05-03',
'DESIGNER', 16, 'F', DATE '1998-04-02', 27740, 2600, 2217);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0210', 'MERLIN M.D', 'D11', '9466976561', DATE '2015-02-10', 'DESIGNER', 17,
'F', DATE '1989-12-28', 18270, 2400, 1462);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0220', 'MARIA JOHN', 'D11', '9466976562', DATE '2016-11-19', 'DESIGNER', 18,
'F', DATE '1995-01-22', 29840, 2600, 2387);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0230', 'VISHALAKSHI V PRABHU', 'D21', '9466976563', DATE '2001-07-21', 'CLERK',
14, 'F', DATE '1972-04-03', 22180, 1400, 1774);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0240', 'ANGEL PAUL', 'D21', '9467976564', DATE '1981-02-01', 'CLERK', 17, 'F',
DATE '1981-02-01', 28760, 1600, 2301);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0250', 'RIYA TONEY', 'D21', '9566976565', DATE '2005-10-23', 'CLERK', 15, 'F',
DATE '1986-09-16', 19180, 1400, 1534);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0260', 'PRIYA TOMY', 'D21', '9665976566', DATE '2013-05-03', 'CLERK', 16, 'F',
DATE '1989-08-09', 17250, 1300, 1380);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0270', 'ARYAMOL ASOKAN', 'D21', '9764976567', DATE '2015-02-10', 'CLERK', 15,
'F', DATE '1991-03-31', 27380, 1500, 2190);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0280', 'GEO GEORGE', 'E11', '9863976568', DATE '2016-11-19', 'OPERATOR', 17,
'M', DATE '1987-01-31', 26250, 1500, 2100);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0290', 'JIMMY THOMSON', 'E11', '9962976569', DATE '2001-07-21', 'OPERATOR', 12,
'M', DATE '1982-05-13', 15340, 1300, 1227);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)

```

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VALUES ('E0300', 'ALAN PAYYAPPILLY', 'E11', '9466976570', DATE '2001-09-21', 'OPERATOR',
14, 'M', DATE '1977-08-31', 17750, 1400, 1420);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0310', 'BEN PETER MATHEW', 'E11', '9466976571', DATE '2005-10-23', 'OPERATOR',
12, 'M', DATE '1981-02-23', 15900, 1300, 1272);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0320', 'KRISHNANUNNI S', 'E21', '9466976572', DATE '2007-06-07', 'FIELDREP',
16, 'M', DATE '1982-06-15', 19950, 1400, 1596);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0330', 'AHALYA V A', 'E21', '9466976573', DATE '2005-01-23', 'FIELDREP', 14,
'F', DATE '1987-05-23', 25370, 1500, 2030);
INSERT INTO employee (EmpNo, EmpName, DeptNo, Mobile, Hire_date, Job, Edlevel, gender,
dob, salary, bonus, comm)
VALUES ('E0340', 'ANJALI NAIR', 'E21', '9466976574', DATE '2012-04-04', 'FIELDREP', 14,
'F', DATE '1992-05-22', 20370, 1500, 2030);
-- =====
-- PROJECT TABLE
-- =====
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('AD3100', 'ADMIN SERVICES', 'D01', 'E0010', DATE '2017-06-24', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('AD3110', 'GENERAL AD SYSTEMS', 'D21', 'E0070', DATE '2016-06-23', DATE
'2017-11-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('AD3111', 'PAYROLL PROGRAMMING', 'D21', 'E0230', DATE '2017-06-24', DATE
'2017-11-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('AD3112', 'PERSONNEL PROGRAMMING', 'D21', 'E0250', DATE '2017-06-24', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('AD3113', 'ACCOUNT PROGRAMMING', 'D21', 'E0270', DATE '2017-11-22', DATE
'2019-01-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('IF1000', 'QUERY SERVICES', 'C01', 'E0030', DATE '2017-09-22', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('IF2000', 'USER EDUCATION', 'C01', 'E0030', DATE '2018-05-24', DATE
'2019-05-24');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('MA2100', 'WELD LINE AUTOMATION', 'D01', 'E0010', DATE '2018-04-24', DATE
'2019-04-24');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('MA2110', 'W L PROGRAMMING', 'D11', 'E0060', DATE '2017-06-24', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('MA2111', 'W L PROGRAM DESIGN', 'D11', 'E0020', DATE '2019-06-24', DATE
'2019-10-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('MA2112', 'W L ROBOT DESIGN', 'D11', 'E0150', DATE '2019-02-05', DATE
'2019-11-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('OP1000', 'OPERATION SUPPORT', 'E01', 'E0050', DATE '2017-06-24', DATE
'2018-03-24');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)

```

```

VALUES ('OP1010', 'OPERATION', 'E11', 'E0090', DATE '2017-06-24', DATE '2020-08-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('OP2000', 'GEN SYSTEM SERVICES', 'E01', 'E0050', DATE '2017-06-24', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('OP2010', 'SYSTEMS SUPPORT', 'E21', 'E0100', DATE '2017-06-24', DATE
'2018-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('OP2011', 'SCP SYSTEM SUPPORT', 'E21', 'E0320', DATE '2018-07-25', DATE
'2019-08-22');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('OP2012', 'APPLICATIONS SUPPORT', 'E21', 'E0330', DATE '2017-06-24', DATE
'2021-07-25');
INSERT INTO project (ProjNo, ProjName, DeptNo, ResEmp, StDate, EndDate)
VALUES ('PL2100', 'WELD LINE PLANNING', 'B01', 'E0020', DATE '2016-06-23', DATE
'2020-11-22');
-- =====
-- EMP_PROJ TABLE
-- =====
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3100', 'E0010', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3100', 'E0020', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3100', 'E0060', '0.75');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3100', 'E0140', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3110', 'E0070', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3110', 'E0090', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3110', 'E0100', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3111', 'E0230', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3111', 'E0240', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3112', 'E0250', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3112', 'E0130', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3112', 'E0140', '0.75');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3113', 'E0270', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3113', 'E0110', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('AD3113', 'E0130', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF1000', 'E0030', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF1000', 'E0260', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF2000', 'E0030', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF2000', 'E0250', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF2000', 'E0260', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('IF2000', 'E0270', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2100', 'E0010', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2100', 'E0330', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2110', 'E0060', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2110', 'E0340', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2111', 'E0020', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2111', 'E0120', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2111', 'E0130', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2112', 'E0150', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2112', 'E0200', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('MA2112', 'E0210', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP1000', 'E0050', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP1000', 'E0310', '0.75');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP1010', 'E0090', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP1010', 'E0290', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2000', 'E0050', '1');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2000', 'E0150', '0.75');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2000', 'E0160', '0.25');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2010', 'E0100', '0.5');

```

```
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2010', 'E0150', '0.5');
INSERT INTO Emp_Proj (ProjNo, EmpNo, EmpTime) VALUES ('OP2011', 'E0320', '0.25');
```

Create table TESTEMP with columns EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS:

```
create table TESTEMP(
EMPNO char(10),
LASTNAME varchar2(30),
WORKDEPT char(5),
HIREDATE date,
SALARY number(10,2),
BONUS number(10,2)
);
```

```
SQL> create table TESTEMP(
 2  EMPNO char(10),
 3  LASTNAME varchar2(30),
 4  WORKDEPT char(5),
 5  HIREDATE date,
 6  SALARY number(10,2),
 7  BONUS number(10,2)
 8  );
```

Table created.

```
SQL> |
```

Add new employees Smith, Baker and Thomas to the TESTEMP table:

```
insert into TESTEMP values ('E0111', 'SMITH', 'C01', DATE '2015-06-06', 25000, 0);
insert into TESTEMP values ('E0222', 'BAKER', 'A00', DATE '2015-06-06', 28000, NULL);
insert into TESTEMP values ('E0333', 'THOMAS', 'D11', DATE '2015-06-06', 33000, 0);
```

```
SQL> insert into TESTEMP values ('E0111', 'SMITH', 'C01', DATE '2015-06-06', 25000, 0);
insert into TESTEMP values ('E0222', 'BAKER', 'A00', DATE '2015-06-06', 28000, NULL);
```

1 row created.

```
SQL>
```

1 row created.

```
SQL> insert into TESTEMP values ('E0333', 'THOMAS', 'D11', DATE '2015-06-06', 33000, 0);
```

1 row created.

```
SQL> |
```

Insert into TESTEMP by copying columns from EMPLOYEE for employees with salary <= 25000:

```
insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
select EmpNo, EmpName, DeptNo, Hire_date, salary, bonus
from employee
where salary <= 25000;
```

```
SQL> insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
  2  select EmpNo, EmpName, DeptNo, Hire_date, salary, bonus
  3  from employee
  4  where salary <= 25000;

14 rows created.

SQL> |
```

Mr. Smith receives a bonus of 500:

```
update TESTEMP
set BONUS = 500
where EMPNO = 'E0111';
```

```
SQL> update TESTEMP
  2  set BONUS = 500
  3  where EMPNO = 'E0111';

1 row updated.

SQL>
```

All employees in department C01 receive a salary increase of 1,000:

```
update TESTEMP
set SALARY = SALARY + 1000
where WORKDEPT = 'C01';
```

```
SQL> update TESTEMP
  2  set SALARY = SALARY + 1000
  3  where WORKDEPT = 'C01';

2 rows updated.

SQL> |
```

Mr. Smith leaves the company:

```
delete from TESTEMP
where EMPNO = 'E0111';
```

```
SQL> delete from TESTEMP
  2  where EMPNO = 'E0111';

1 row deleted.

SQL> |
```

Insert into TESTEMP by copying columns from EMPLOYEE for employees with salary > 25000:

```
insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
select EmpNo, EmpName, DeptNo, Hire_date, salary, bonus
from employee
where salary > 25000;
```

```
SQL> insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
  2  select EmpNo, EmpName, DeptNo, Hire_date, salary, bonus
  3  from employee
  4  where salary > 25000;

18 rows created.

SQL>
SQL>
```

Ammu Anna Pius is assigned to department E01:

```
update employee
set DeptNo = 'E01'
where EmpName = 'AMMU ANNA PIUS';
```

```
SQL>
SQL> update employee
  2  set DeptNo = 'E01'
  3  where EmpName = 'AMMU ANNA PIUS';

0 rows updated.

SQL>
```

Mrs. Brown joins the company (E0360, D01, hired today, salary 45000, bonus unknown):

```
insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
values ('E0360', 'BROWN', 'D01', SYSDATE, 45000, NULL);
```

```
SQL> insert into TESTEMP (EMPNO, LASTNAME, WORKDEPT, HIREDATE, SALARY, BONUS)
  2  values ('E0360', 'BROWN', 'D01', SYSDATE, 45000, NULL);

1 row created.

SQL>
```

Delete all rows from the *TESTEMP* table and drop the *TESTEMP* table:

```
delete from TESTEMP;
```

```
drop table TESTEMP;
```

```
SQL> delete from TESTEMP;
```

```
35 rows deleted.
```

```
SQL>
```

```
SQL> drop table TESTEMP;
```

```
Table dropped.
```

```
SQL> |
```